MIP Formulations for Delete-Free AI Planning

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Outline

BACKGROUND AND PRELIMINARY WORK + RESULTS

IMPROVEMENTS TO LM-CUT AND USING IT AS SEPARATOR + RESULTS

Delete-Free Planning : Background & Notation

$$\Pi = \langle P, A, s, t, \mathrm{cost} \rangle$$

- P: Set of boolean variables (facts)
- A: Set of actions, each with
  - $\mathrm{pre}(a) \subseteq P$: Preconditions, facts to be true for the action to be applicable
  - $\mathrm{add}(a) \subseteq P$: Add effects, facts achieved by applying the action
- $s, t \in P$: Initial and Goal facts
- $\mathrm{cost} : A \rightarrow \mathbb{R}_0^+$: Non-negative cost function

Objective: finding the sequence of actions that achieves t, starting from s, that minimizes total cost h^+

Landmarks and LM-Cut

- A disjunctive landmark $L \subseteq A$ is a set of actions such that every plan has at least one action from L
  - Delete-Free planning is equivalent to solving an hitting set problem over all its landmarks
- LM-Cut is a procedure that computes h^{LM}, a relaxation of h^{+}:
  - Iteratively: computes a landmark, look up the lowest cost of the actions in such landmark, reduce those action's costs by that amount, repeat
  - h^{LM} is the sum of the costs computed at each iteration

Bonet & Castillo, "A Complete Algorithm for Generating Landmarks", 2011
Helmert & Domshlak, "Landmarks, critical paths and abstractions: what's the difference anyway?", 2009

MIP Formulation : Basic model

$$\min \sum_{a \in A} \mathrm{cost}(a) x_a$$

$$\sum_{a \in A_{\to q}} z_{a,q} = y_q \qquad \forall q \in P$$

$$\sum_{a \in A_{p \to q}} z_{a,q} \leq y_p \qquad \forall p, q \in P$$

$$z_{a,q} \leq x_a \qquad \forall a \in A, \forall q \in \mathrm{add}(a)$$

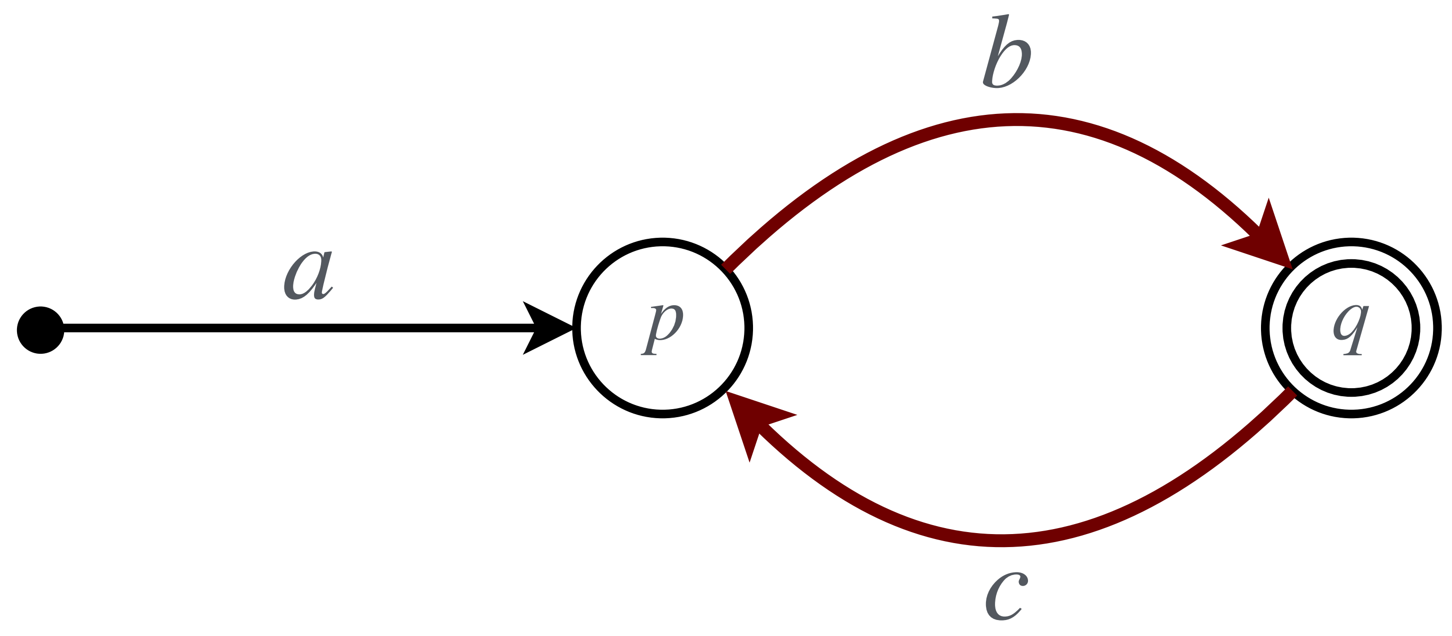
$$y_s, y_t = 1$$

$$x_a, z_{a,q}, y_p \in \{0,1\} \qquad \forall a \in A, \forall q \in \mathrm{add}(a), \forall p \in P$$

Rankooh & Rintanen, "Efficient Computation and Informative Estimation of h+ by Integer and Linear Programming", 2022

MIP Formulation : Acyclicity

- This is not a complete MIP formulation
- We are missing causal acyclicity



- Vertex Elimination: Previous state of the art used Vertex Elimination graphs to encode precedence (through the transitive closure of the causal relation graph of the task)
- Landmark Constraints: Our approach works by adding to the base model the landmarks violated by its solutions

Haslum, Slaney & Thiebaux, "Minimal Landmarks for Optimal Delete-Free Planning", 2012
Rankooh & Rintanen, "Efficient computation and informative estimation of h+ by integer and linear programming", 2022
Salvagnin & Zanella, "MIP Formulations for Delete-Free AI Planning", 2026

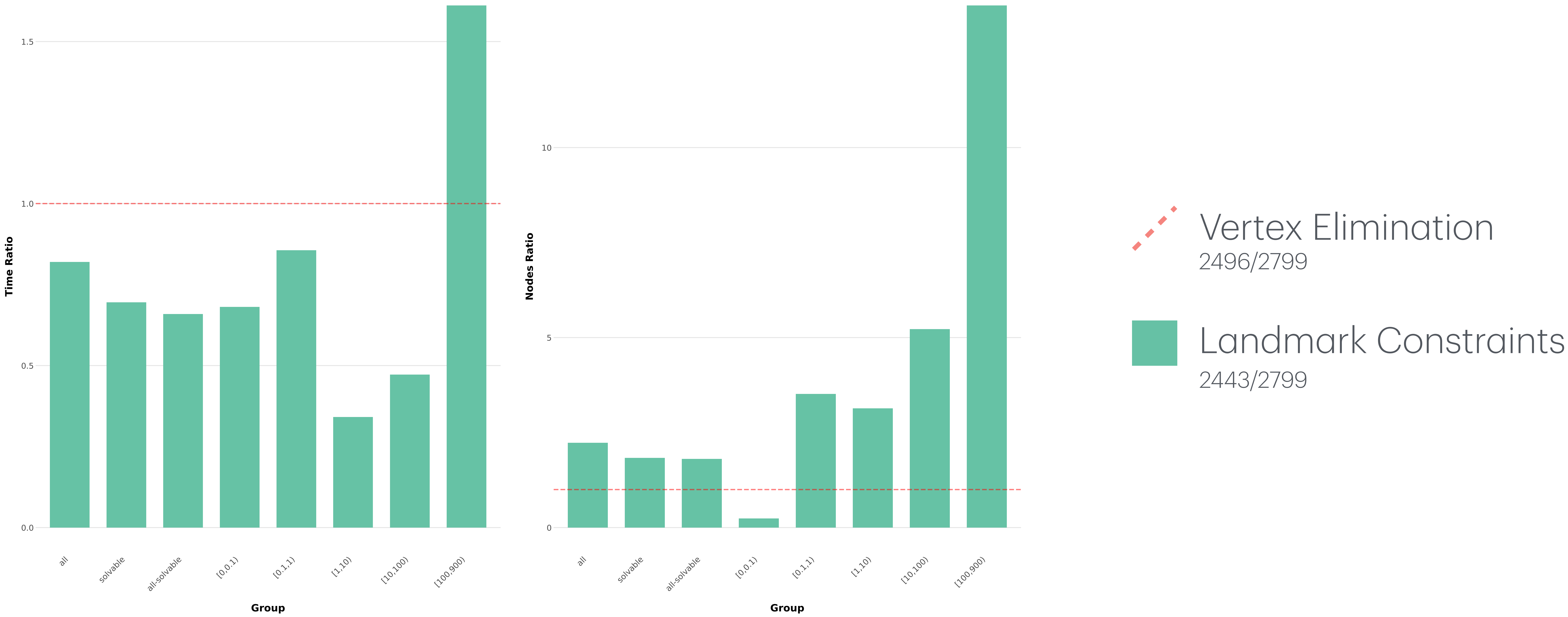
Warm Start

- Primal Heuristic
  - Greedy algorithm
  - Lookahead strategy based on the h^{add} heuristic (a relaxation of h^{+}) to evaluate different actions

- LM-cut initialization
  - We compute LM-cut and use its landmarks as an initial set of constraints: $\sum_{a \in L} x_a \geq 1$
  - Repeated multiple times to take advantage of it's degrees of freedom

Bonet & Geffner, "Planning as heuristic search", 2001
Lauer & Fickert, "Beating LM-cut with LM-cut: Quick cutting and practical tie breaking for the precondition choice function", 2020

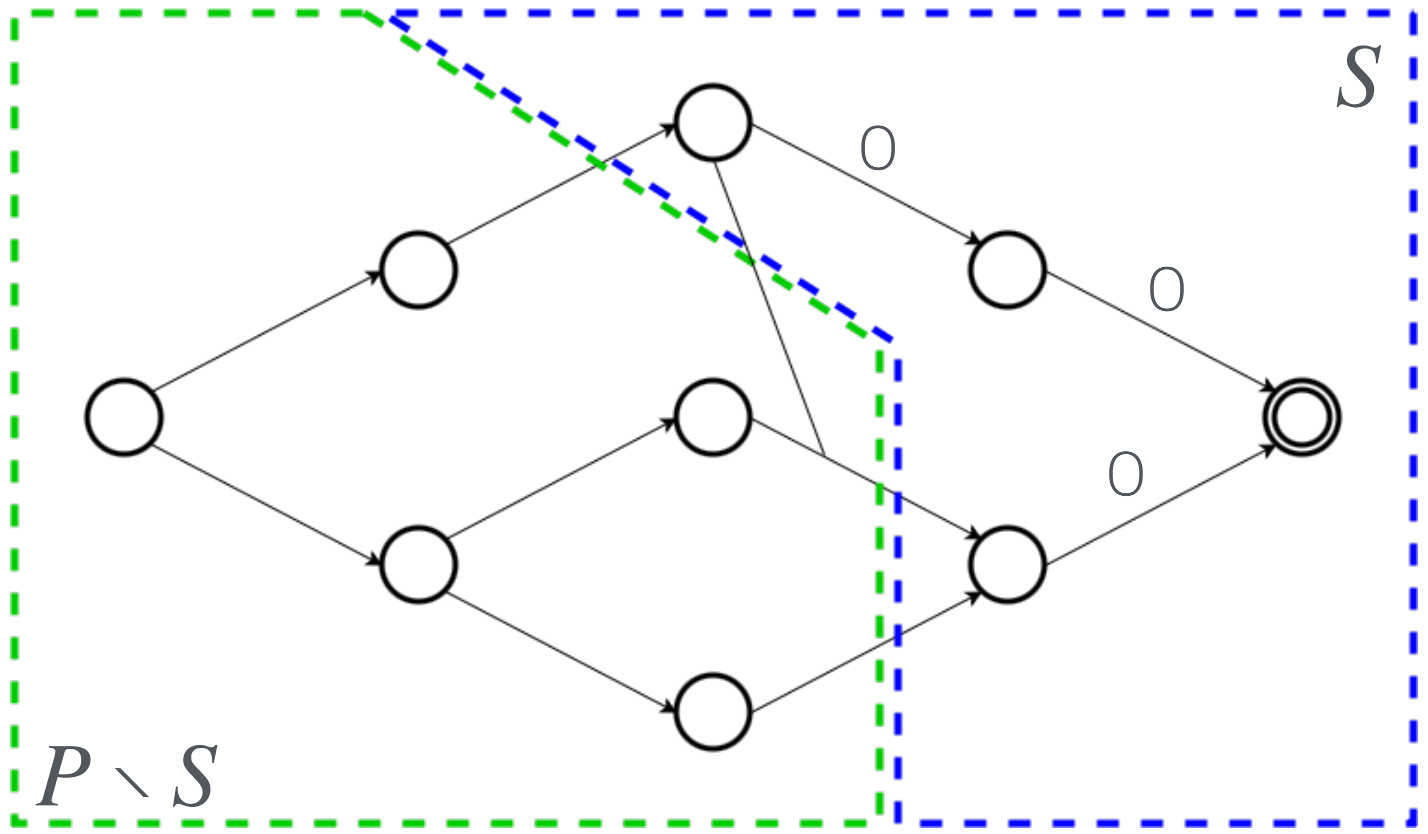
Results: current baseline



time limit: 15m - memory limit (CPLEX): 4GB

Improving LM-Cut

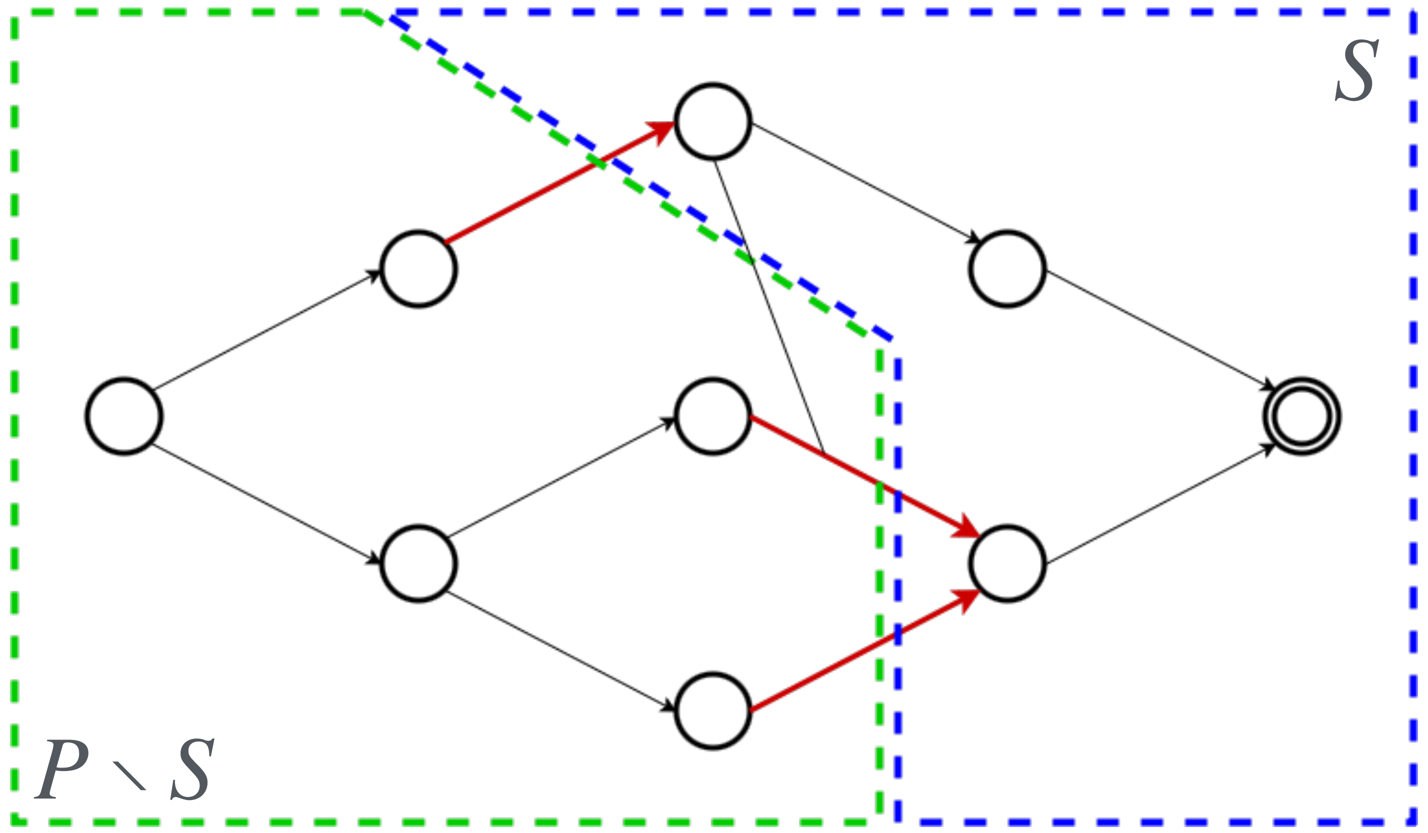
- LM-Cut, at each iteration, builds a justification graph: let $D(a)$ be the precondition with maximum $h^{\max}$ (a further relaxation of h^+) of action a; the graph is then constructed as $G = (P, E = \{(D(a), q) \mid a \in A, q \in \mathrm{add}(a)\})$ (an edge connecting $D(a)$ to each of as effects)
- Partition the facts in goal section S and pre-goal $P \setminus S$, such that S is the set of facts from which t is reachable through a 0-cost path in G

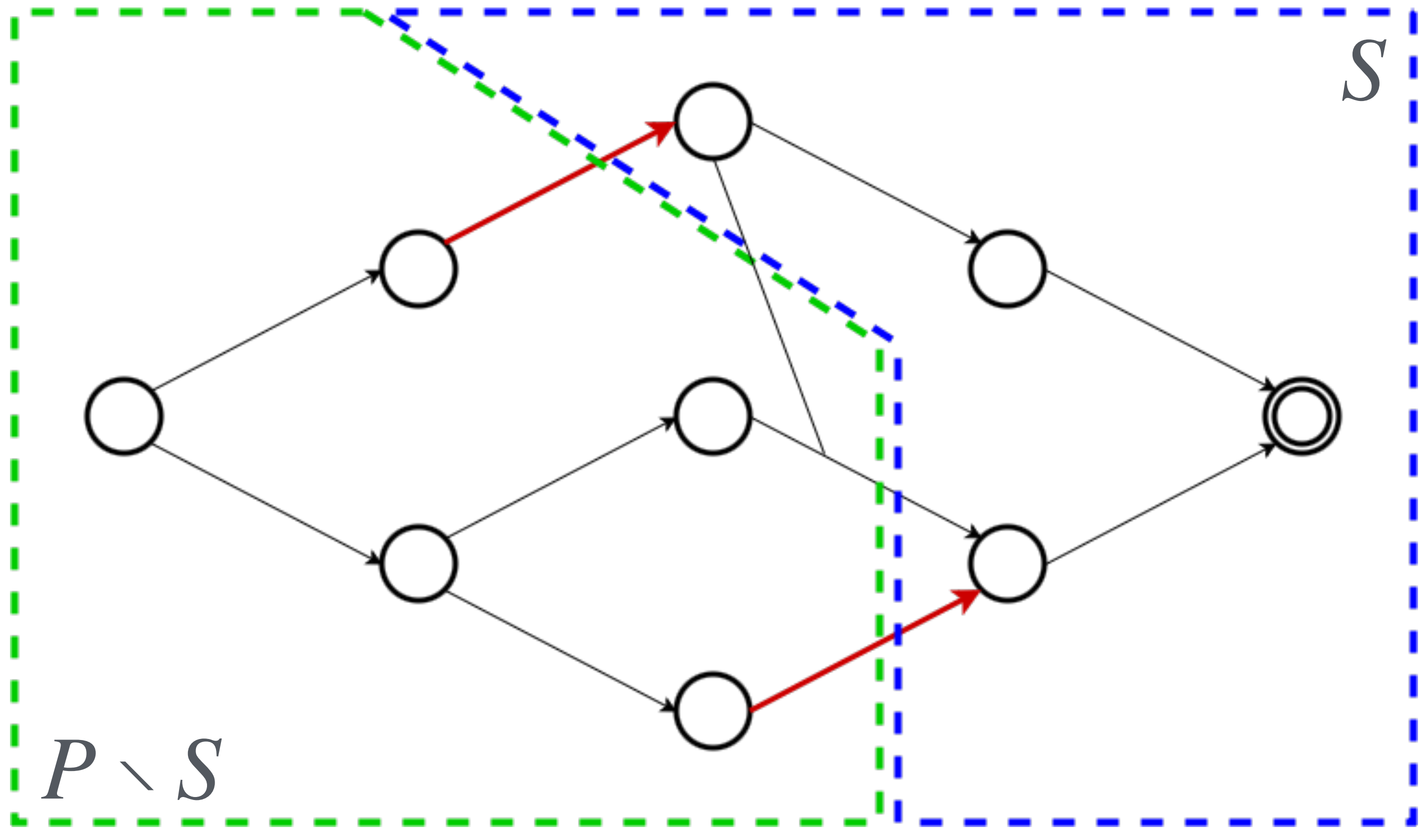


Helmert & Domshlak, *"Landmarks, critical paths and abstractions: what's the difference anyway?"*, 2009

Improving LM-Cut

- The landmark is then computed as the set of actions crossing the partition:
  $L = \{a \in A \mid D(a) \in (P \setminus S), \mathrm{add}(a) \cap S \neq \emptyset\}$
- L however is not minimal (there might exist a smaller landmark $L' \subset L$):
  - Greedy approach: $L = \{a \in A \mid \mathrm{pre}(a) \subseteq (P \setminus S), \mathrm{add}(a) \cap S \neq \emptyset\}$
  - Extensive approach: remove all actions in L such that it's still is a valid landmark





Haslum, Slaney & Thiebaux, "Minimal Landmarks for Optimal Delete-Free Planning", 2012

LM-Cut as a cut separator

- By setting the initial cost of a set of actions to 0:
  - The landmarks obtained are still valid, as they are computed as s-t cuts in G
  - Those actions won't appear in any resulting landmark
  - It's as if their cost had been reduced to 0 in "previous iterations"

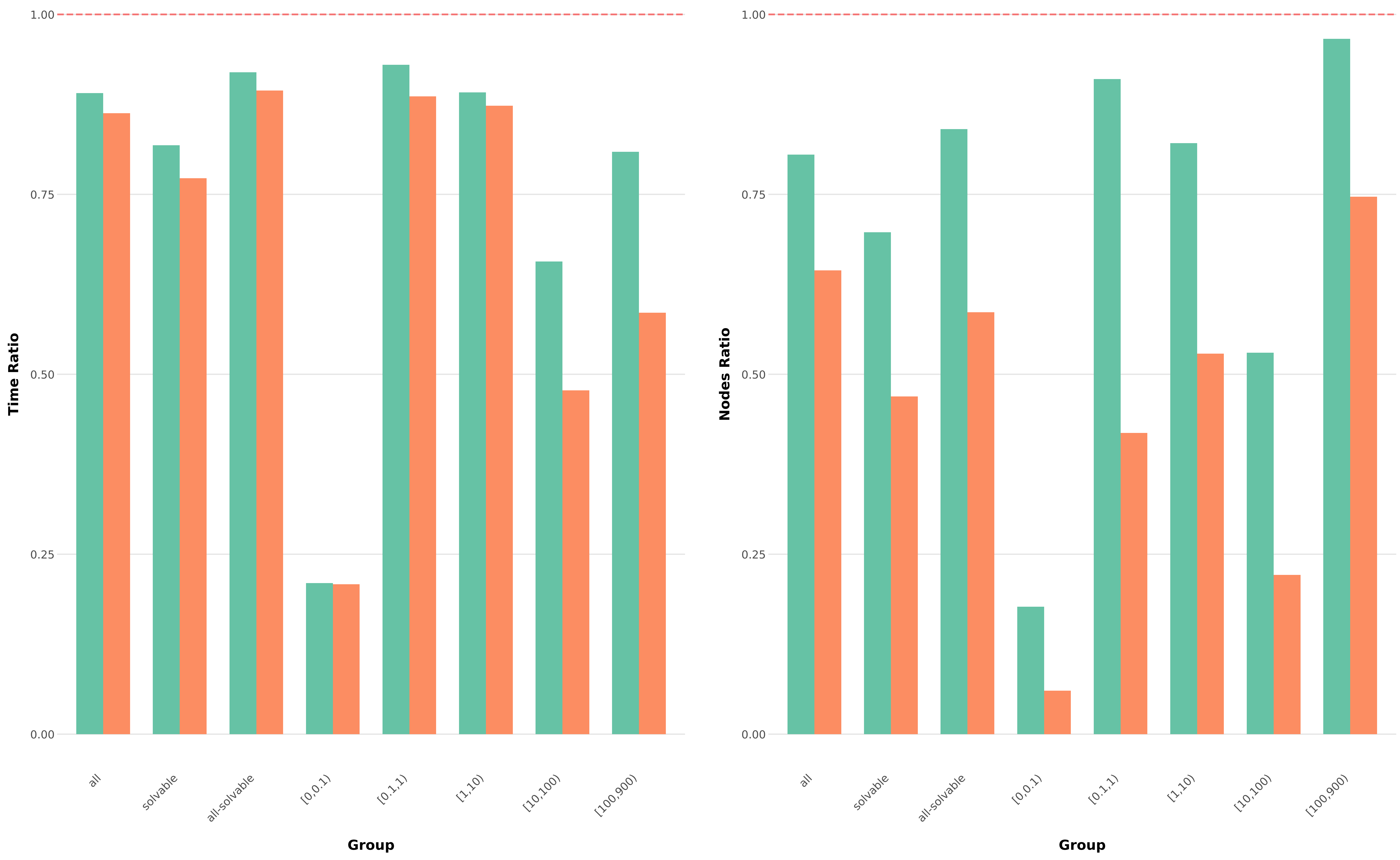
- We can use LM-Cut as an integer solution separator by setting the initial cost of used actions to 0: all the produced landmarks will then be violating the integer solution as they are all valid landmarks composed only by unused actions

LM-Cut as a cut separator

- Deciding wether a fractional solution violates a landmark is an NP-Hard problem

- Given a fractional solution $\tilde{x}$, we generalize the initial cost assigned to each action: $\tilde{c}(a) = (1 - \tilde{x}_a)\, c(a)$.
- Unlike for integer solutions, there's no guarantee that the landmarks produced will be violated by $\tilde{x}$

Results: improving LM-Cut



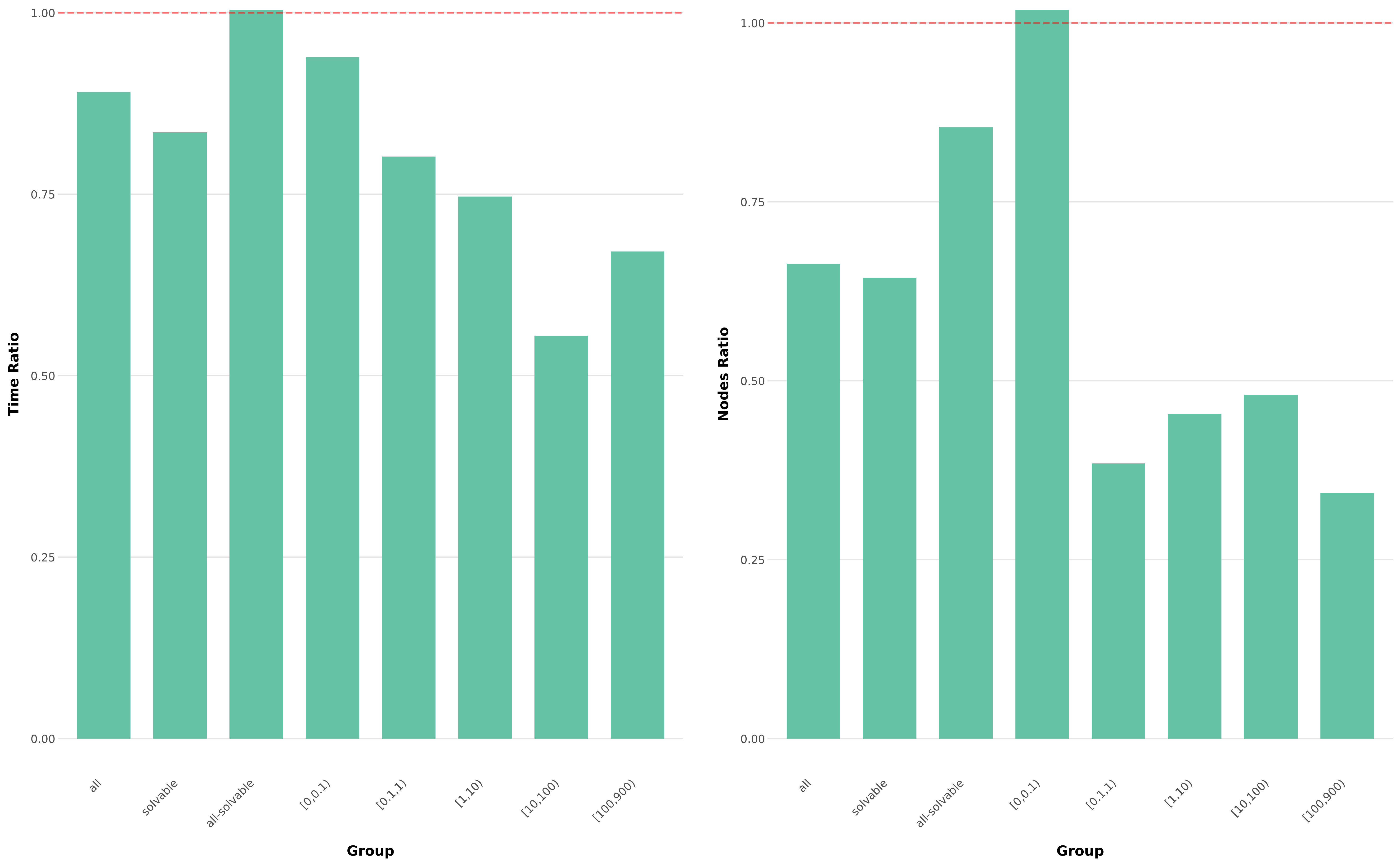
LM
2443/2799

LM + LM-Cut improved greedily
2464/2799

LM + LM-Cut improved extensively
2482/2799

time limit: 15m - memory limit (CPLEX): 4GB

Results: LM-Cut as integer separator



LM (best)

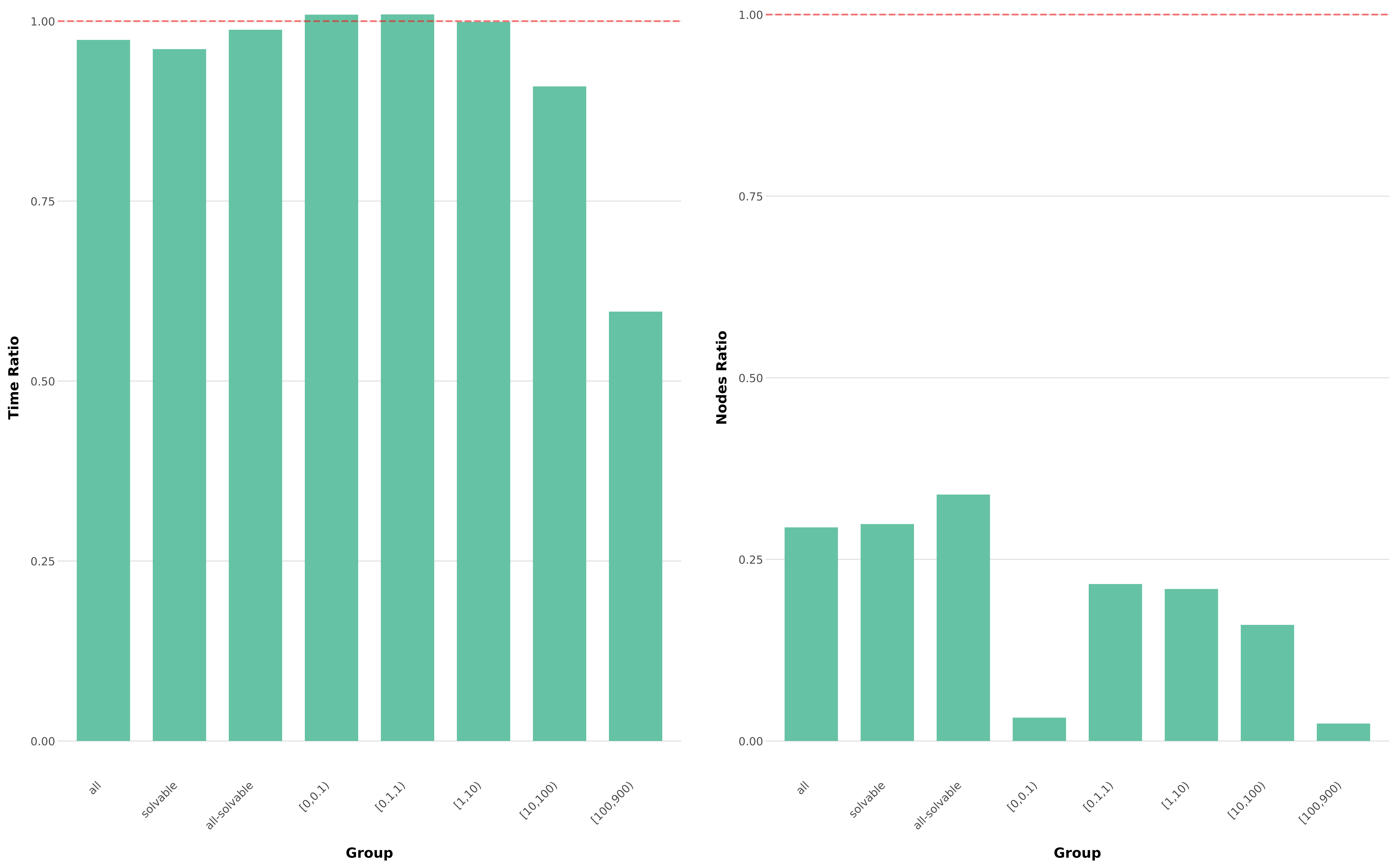
2482/2799

LM-Cut (best) as integer separator

2529/2799

time limit: 15m - memory limit (CPLEX): 4GB

Results: LM-Cut as fractional separator

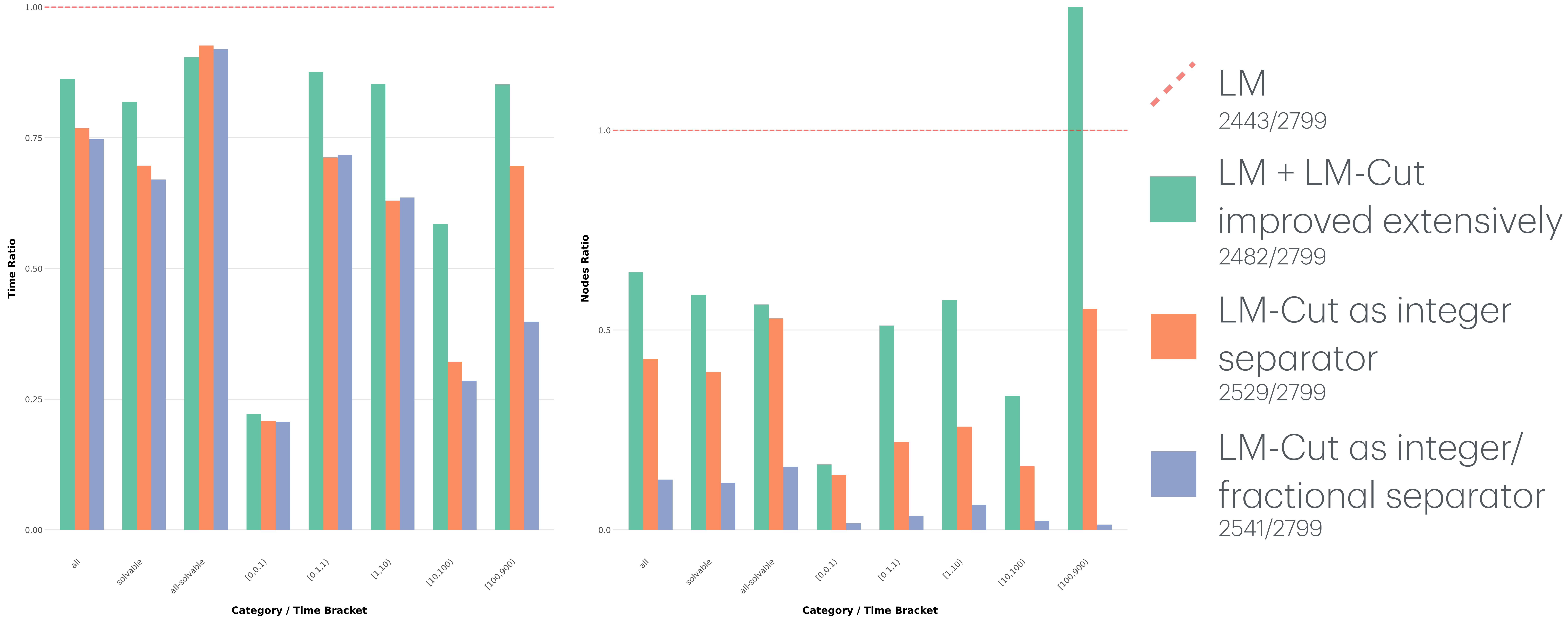


LM-Cut as integer separator
2529/2799

LM-Cut as integer/ fractional separator
2541/2799

time limit: 15m - memory limit (CPLEX): 4GB

Results: overall improvements



time limit: 15m - memory limit (CPLEX): 4GB

Thanks for your attention!

If you have any questions, feel free to contact me: ***matteo.zanella.3@phd.unipd.it***

Code and slides available at: https://github.com/Zanzibarr/MIPxHPLUS/tree/CPAIOR26